

OKAFOR PATRICK CHIBUIKE

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PROFESSIONAL SUMMARY

Full stack, and Software Engineer with 5+ years of building high-performance production systems. Expert in backend optimisation with proven latency reductions and uptime improvements. Proficient in Node.js, Python, Rust, TypeScript, and system-level programming. Specialised in Solana blockchain development, REST APIs, microservices, and real-time systems.

PROFESSIONAL EXPERIENCE

Lead Backend & IoT Cloud Engineer – AllinzucolSmart Lab | August 2026 – Present

Cloud-Connected Hospitality & IoT Operations Platform (NestJS/TypeScript, Bun, AWS, MQTT, PostgreSQL)

- Architected and deployed an enterprise IoT hospitality management platform on AWS (ECS Fargate, AWS IoT Core, RDS PostgreSQL, ElastiCache Redis) using NestJS, TypeScript, and Bun.
- Engineered a high-throughput IoT ingestion pipeline over MQTT with mutual TLS (mTLS) via AWS IoT Core, normalising raw telemetry (PIR motion, door contacts, temperature/humidity, relays) into decoupled event streams with < 50 ms processing latency.
- Designed a safety-critical Emergency Alert System for real-time hazard detection (smoke, fire, panic, intrusion, water leak, device fault) featuring automated multi-tier escalation and sub-second WebSocket broadcast via [Socket.IO](#).
- Implemented deterministic Finite State Machines (FSMs) governing room lifecycle states and housekeeping workflows (Pending → Assigned → In-Progress → Inspection → Vacant Clean), eliminating race conditions and automating turnover workflows.
- Built granular **6-tier Role-Based Access Control (RBAC)**, JWT authentication with refresh token rotation, per-device hardware provisioning, and immutable RFC 7807 audit logging.
- Enforced end-to-end runtime type safety with **TypeBox** validation across HTTP APIs and **MQTT** payloads with zero `any` tolerance, maintaining sub-15ms p95 API response times.
- Developed an operational analytics engine aggregating real-time room turnaround metrics, staff cleaning efficiency, and device health telemetry for supervisory dashboards.
- Achieved 95%+ test coverage across 200+ unit and integration tests using Jest and simulated MQTT hardware test harnesses, validating offline heartbeats, state transitions, and edge failures.

Lead Backend and Smart Contract Engineer – Zephyr Protocol | Jan 2026 – Present

Solana Copy Trading Protocol (Rust/Anchor + Node.js/TypeScript)

- Architected a non-custodial vault system using Anchor PDAs for transparent trade execution and automated mirror trading
- Optimized infrastructure latency across 3 regions, achieving 16x reduction (14ms to 0.88ms RPC latency)
- Designed a 5-PDA on-chain governance model enabling parameter adjustments without program redeployment
- Implemented 5-tier dynamic revenue model with auto-scaling split ratios based on on-chain volume, PnL, and AUM
- Integrated Pyth/Switchboard oracles and Jupiter V6 swap routing for automated stop-losses and take-profits
- Built real-time indexer monitoring on-chain trades with sub-block latency for mirror-trade execution
- Developed analytics pipeline computing Sharpe Ratio, rolling AUM, and ROI using PostgreSQL and Prisma
- Implemented wallet-based auth (Ed25519 + JWT) and verifiable leaderboard ranking traders by on-chain metrics

Backend Engineer – KENET Health | Dec 2024 – Mar 2025

Telemedicine Platform (Node.js/Express + MongoDB)

- Designed scalable backend architecture supporting a growing user base with seamless frontend integration
- Improved onboarding flow, reducing new user access time by 40%
- Integrated third-party APIs for payment processing and health data management
- Optimised backend performance, achieving a 30% reduction in API response times
- Documented API endpoints using Swagger for developer onboarding

PROJECTS

Okaform – Solana-Native Survey Platform

Solana, Rust, Anchor, NestJS, MongoDB, React, TypeScript, Tailwind CSS

- Built full Anchor program with 6 instructions and 25+ integration tests for survey lifecycle management

- Implemented push-based reward distribution using ALTs for batched transfers of 25-30 recipients per transaction
- Designed wallet-based authentication with SIWS, JWT refresh rotation, and Ed25519 signature verification
- Built a sybil filtering system checking wallet age, SOL balance, and funding-graph analysis
- Developed NestJS backend with TypeBox validation, custom exceptions, and structured logging (78 unit tests)
- Created React frontend with drag-and-drop form builder, live preview, and wallet-gated survey filling
- Integrated on-chain escrow with automatic reward distribution weighted by reputation scores

High-Performance TCP Server

Python, SSL/TLS, Multithreading, Systemd

- Architected multithreaded TCP server handling 100+ concurrent clients with sub-millisecond response times
- Engineered 10 search algorithms achieving 7-15ms on 250K-line files (2.7-5.7x faster than required)
- Implemented SSL/TLS encryption, rate limiting, connection management, and DoS mitigation
- Achieved 100% test coverage with unit, integration, and SSL handshake validation

MemeStream Agent (TypeScript)

TypeScript, AI engineer, Google Gemini SDK, ExpressJS, Telegram-bot

- Architected an AI-powered Telegram bot using Node.js, TypeScript, and Docker, serving meme discovery to an active user base
- Engineered a queue-based processing system with BullMQ and Redis, achieving 99.5% uptime for asynchronous task management
- Integrated Google Gemini AI with function calling and streaming for context-aware meme recommendations
- Implemented an automated web scraping pipeline with Playwright, maintaining a dynamic database of 100+ meme templates
- Established CI/CD pipeline with GitHub Actions and PM2 for zero-downtime deployments

ProjectRUGGUARD – Twitter Risk Analysis Bot

Python, Twitter API, NLP, Caching

- Architected a Twitter bot providing real-time risk analysis for 2,000+ users

- Engineered a heuristic engine analysing 10+ trust signals, including account age and follower ratios
- Implemented multi-layer caching reducing redundant API calls by 60%
- Built automated report generation with colour-coded trust scoring via Twitter API

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Rust, C, Bash

Frameworks: Node.js, NestJS, Express, Django, Actix, Flask, Anchor (Solana)

Databases: MongoDB, PostgreSQL, MySQL, Redis, SQLite

APIs/Protocols: REST, gRPC, WebSocket, BullMQ, RabbitMQ, MQTT (TLS/mTLS), Socket.IO

Blockchain: Solana, Anchor, Rust, PDAs, Wallet Integration

Cloud/DevOps: Docker, GitHub Actions, Grafana, Prometheus, ELK Stack, Nginx, AWS, Hetzner

Testing: Jest, Pytest, Unit Testing, Integration Testing

EDUCATION

B.Sc. Biotechnology – Ebonyi State University, Nigeria

CERTIFICATIONS

- Solana/Rust Developer – Encode (Nov 2025)
- ALX AI Starter Kit Program (2025)
- ALX Software Engineering Program (2024)
- FullStack Certified Professional – Zuri X I4G (2022)